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Deposit Modeling

Corporate Treasury Quantitative Analytics
Case Study

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Deposit Modeling Is Like A Reliable Weather Forecast Before Sailing

- Predict behavior of customer funds under stress
- Show that can absorb losses without collapsing
- Calculate liquidity needs and capital buffers, under economic scenarios (downturns or market shocks)
- Align corporate strategies with realistic deposit flows
- Not to overestimate cash for growth or underestimate risks in a crisis

Understanding Deposits Behavior Is Crucial: Customer Loyalty, Market Rates, And Trust

Non-Interest-Bearing Deposits

- Super stable, everyday transactions, less sensitive to rates
- Might see outflows if customer confidence drops

Non-Maturity Interest-Bearing Deposits

- More sensitive to rate changes
- Not as sticky as non-interest-bearing deposits

Deposits Make Up The Bulk Of FDIC's Quarterly Banking Profile

Data from FDIC includes:

Historical quarterly deposit data (1991-2023)

- Non-interest-bearing deposits
- Non-maturity Interest-bearing deposits
- Time deposits

Historical economic variables (1976-2023)

- Real/Nominal GDP and growth rates
- Unemployment rate
- Interest rates (3-month Treasury, 5-year, 10-year, BBB corporate, mortgage, prime)
- Inflation metrics (CPI)
- Stock market indices
- House price indices

Four economic scenarios for forecasting (2024-2027)

- Supervisory Baseline
- Supervisory Severely Adverse
- Exploratory Conditions A
- Exploratory Conditions B

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Non-Interest-Bearing Deposits

Trend

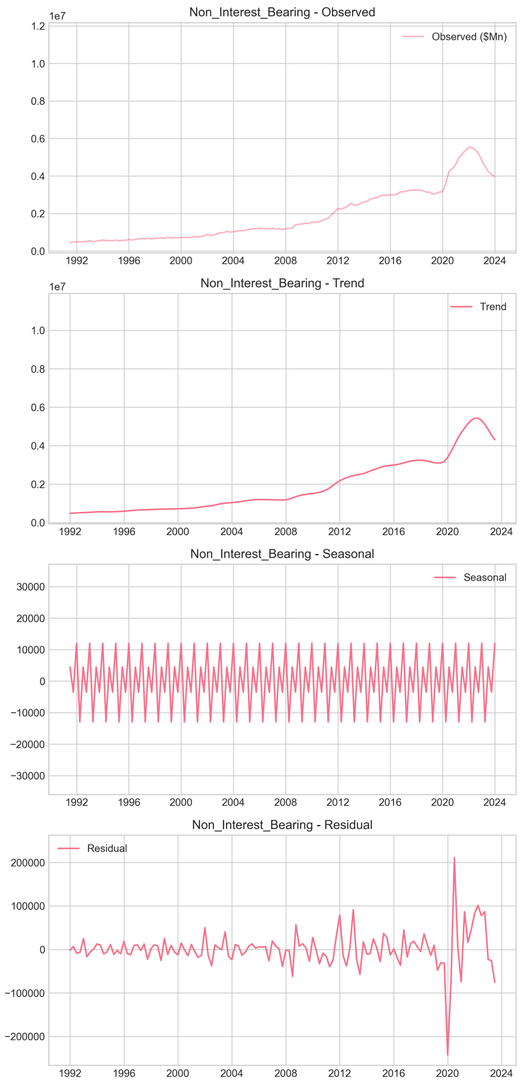
- Strong upward trend from ~$1T (2008) to ~$5T (2020)
- Massive acceleration during COVID-19 period (2020-2021)
- Post-2021 plateau with slight decline - normalization after stimulus

Seasonality

- Clear quarterly seasonality with amplitude ~$10-15B
- Q4/Q1 peaks - likely year-end business cash accumulation and tax-related flows
- Mid-year dips in Q2/Q3 - consistent pattern across decades

Residuals

- Large residual spikes around 2008 (GFC) and 2020-2021 (COVID)
- Generally well-behaved outside crisis periods
- Recent volatility suggests ongoing structural shifts



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Non-Maturity Interest-Bearing Deposits

Trend

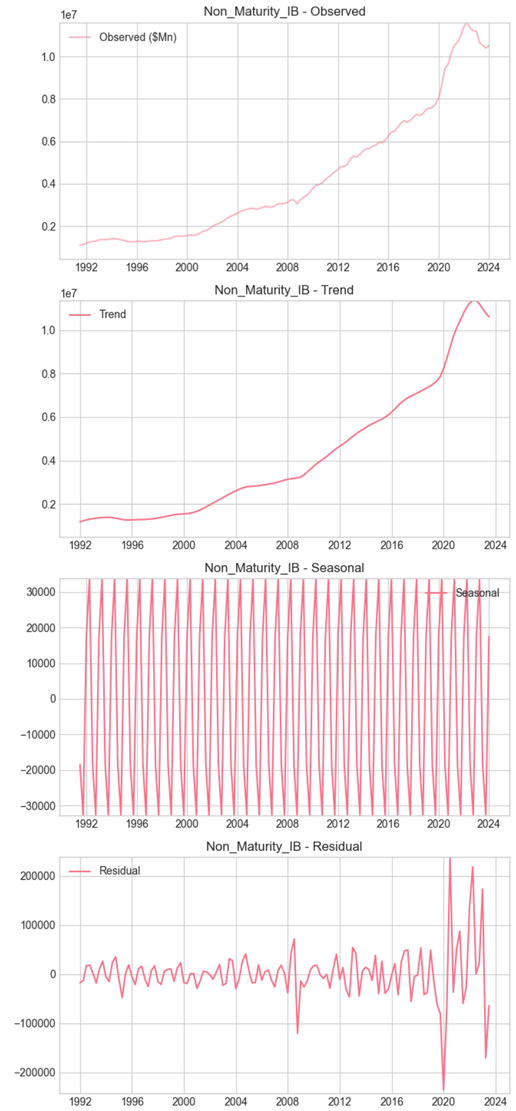
- Surged from ~$2T (2008) to ~$11T (2021), steepest growth among deposit types
- Fueled by low rates and stimulus during COVID-19
- Recent leveling off, but balances remain historically high

Seasonality

- Sharp quarterly swings, ~$30B amplitude
- Similar Q4/Q1 peak pattern to non-interest-bearing, but more volatile
- Seasonality has grown with deposit base - larger absolute swings over time

Residuals

- Massive residuals during COVID period due to stimulus driven inflows
- Post-2020 elevated residuals indicate persistent structural shifts
- Pre-COVID models fit better; recent volatility challenges predictions



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Time Deposits

Trend

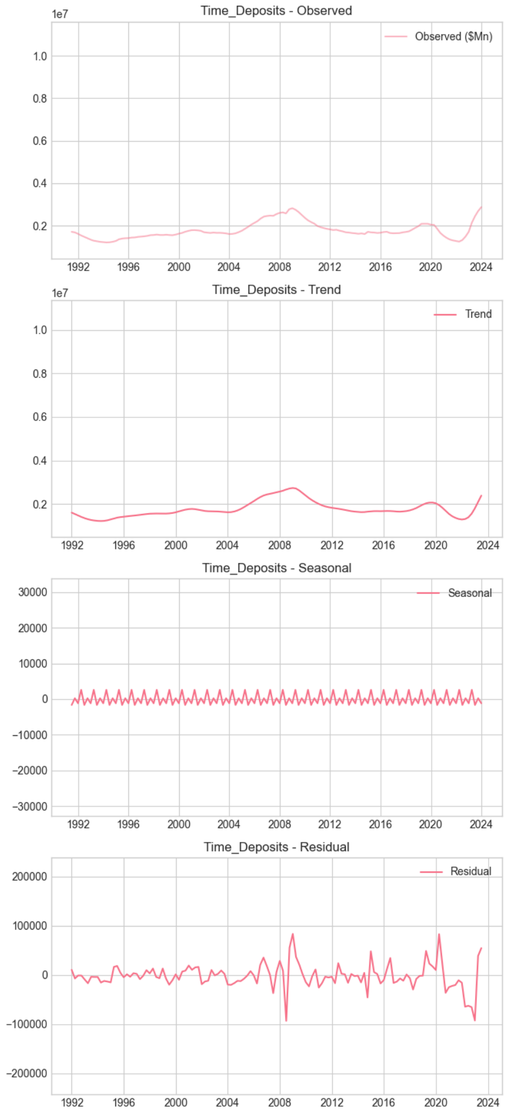
- Peaked at ~$3T (2009), declined to ~$1.5T (2016-2019)
- Recent uptick starting 2022 - likely due to rising interest rates making CDs attractive again
- Steadiest trend among deposit types, with minimal volatility

Seasonality

- Minimal seasonality - most stable seasonal component
- Smaller amplitude (~$5-10B) reflects their fixed-maturity nature
- Less sensitive to quarterly business/tax cycles

Residuals

- Most stable residuals throughout most periods
- Smaller crisis impacts - more predictable behavior
- Recent uptick residuals reflect interest rate regime change



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Stationarity tests

Test	Null Hypothesis	Alternative	Key Characteristics	Interpretation
Augmented Dickey-Fuller (ADF)	Unit root exists (non-stationary)	Stationary	- Tests for unit root - Includes trend/drift terms - p < 0.05 → reject null → stationary	Low p-value = stationary
Kwiatkowski-Phillips-Schmidt-Shin (KPSS)	Series is stationary	Unit root exists	- Reverse logic of ADF - Tests stationarity as null - p > 0.05 → fail to reject → stationary	High p-value = stationary
Phillips-Perron (PP)	Unit root exists	Stationary	- Non-parametric version of ADF - Corrects for serial correlation - More robust to heteroscedasticity	Low p-value = stationary

First Differencing of Deposits	Test	Statistic	P-value	5% Critical Value	Conclusion	Status
Non-Interest-Bearing Deposits	ADF	-2.59	0.094	-2.89	Borderline non-stationary	⚠️
	KPSS	0.093	>0.10	0.463	Strongly stationary	✅
	PP	-5.06	<0.001	-2.88	Strongly stationary	✅
Non-Maturity Interest-Bearing Deposits	ADF	-2.69	0.076	-2.89	Borderline non-stationary	⚠️
	KPSS	0.452	0.055	0.463	Stationary	✅
	PP	-6.24	<0.001	-2.88	Strongly stationary	✅
Time Deposits	ADF	-3.90	0.002	-2.88	Strongly stationary	✅
	KPSS	0.122	>0.10	0.463	Strongly stationary	✅
	PP	-3.24	0.018	-2.88	Stationary	✅

Non-Interest-Bearing Deposits

AR(1) Captures Momentum

- PACF lag-1 spike (~0.6) confirms non-seasonal AR(1), modeling deposit growth persistence

Stationary with Differencing

- First differencing (d=1) ensures stationarity, supported by ACF decay

Seasonal AR(2) for Quarterly Peaks

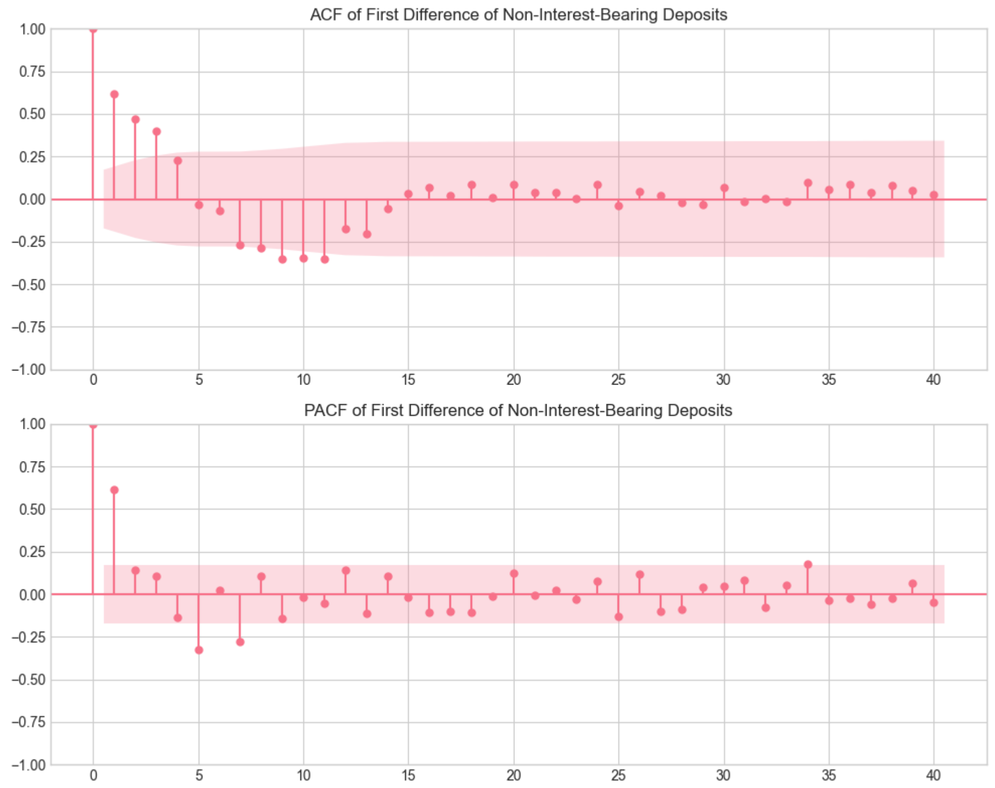
- AR(2) at lag 4/8 models Q4/Q1 tax and business cash peaks

No MA Needed

- Rapid ACF cutoff supports omitting MA (q=0), keeping model simple

Economic Fit

- Reflects persistent growth and quarterly cash flows



SARIMAX(1,1,0)(2,0,0,4)

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Non-Maturity Interest-Bearing Deposits

AR(3) for Persistence

- PACF spikes at lags 1-3 (~0.55, ~0.25, 0.25) with sharp cutoff confirm AR(3), capturing 3-quarter momentum in deposit growth

MA(1) for Shocks

- ACF's lag-1 spike (~0.55) and exponential decay support MA(1), modeling single-period shock absorption (e.g., rate changes)

Stationarity via Differencing

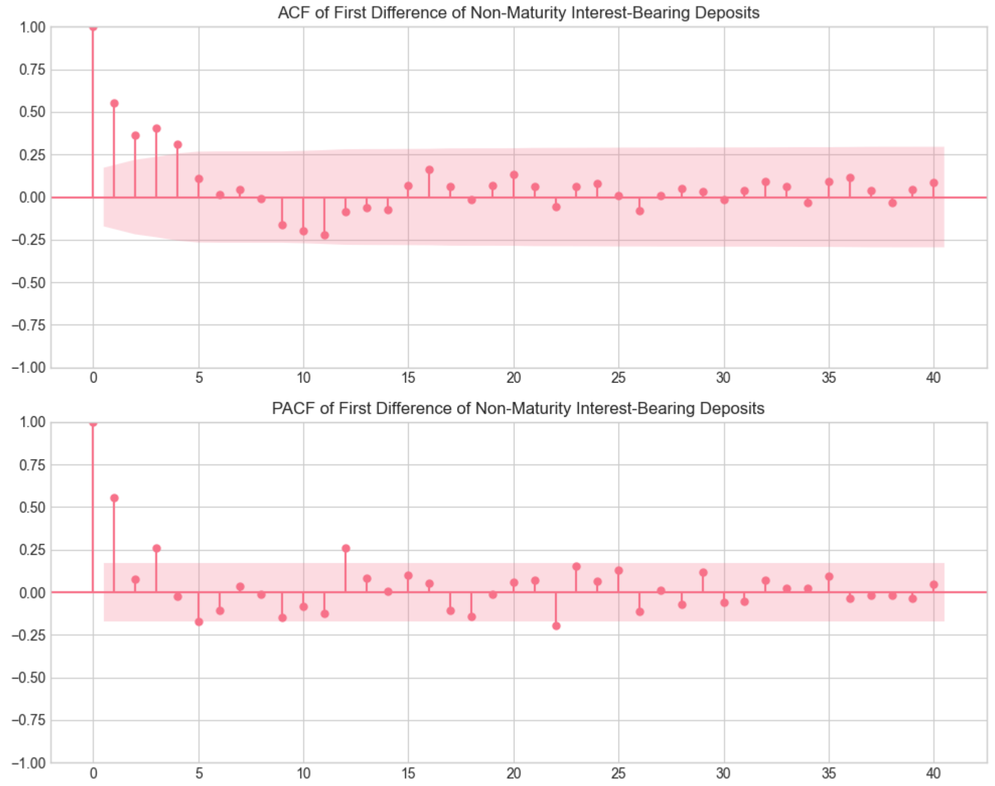
- First differencing (d=1) ensures non-seasonal stationarity; seasonal differencing (D=1) handles quarterly trends.

No Seasonal AR/MA

- No PACF/ACF spikes at lags 4, 8, 12 (P=0, Q=0), avoiding over-parameterization while capturing seasonality

Economic Fit

- Models 3-quarter savings behavior, rapid shock adjustment, and quarterly cycles



SARIMAX(3,1,1)(0,1,0,4)

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Time Deposits

AR(1) for Momentum

- PACF lag-1 spike (~0.75) with sharp cutoff confirms AR(1), capturing quarter-to-quarter deposit flow persistence

Seasonal AR(1) for Cycles

- PACF spike at lag 4 (~0.10) supports SAR(1), modeling annual tax/seasonal patterns

No MA Needed

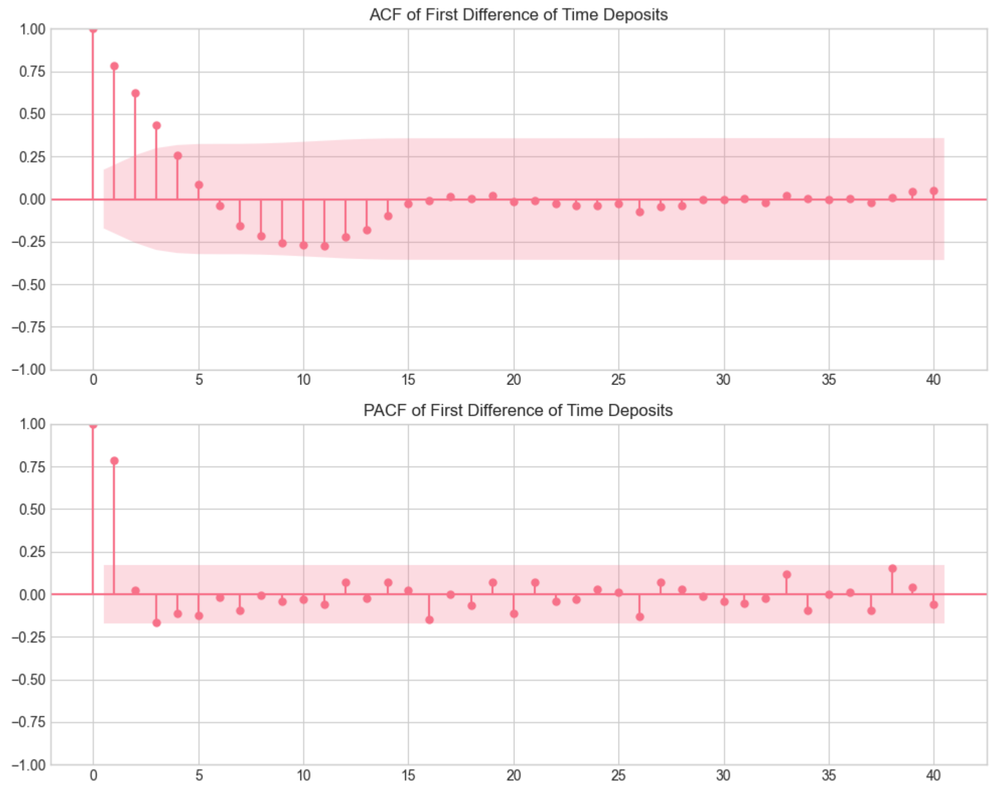
- ACF's gradual decay, not sharp cutoff, rules out MA (q=0, Q=0), keeping model simple

Stationarity Achieved

- First differencing (d=1) ensures stationarity; no seasonal differencing (D=0) needed

Economic Fit

- Reflects stable, momentum-driven CD behavior and quarterly cycles



SARIMAX(1,1,0)(1,0,0,4)

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Deposits Make Up The Bulk Of FDIC's Quarterly Banking Profile

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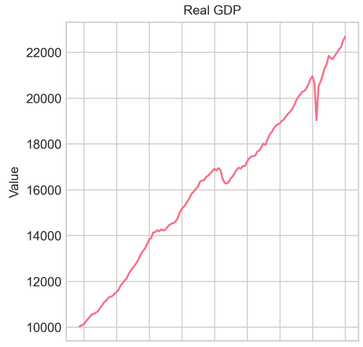
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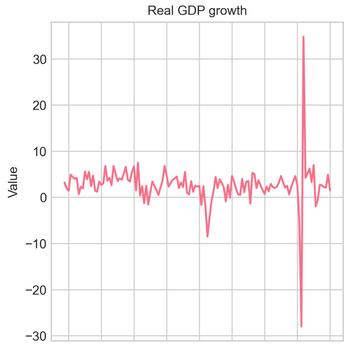
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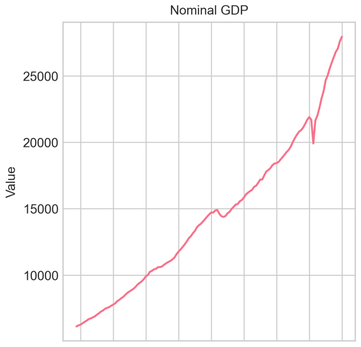
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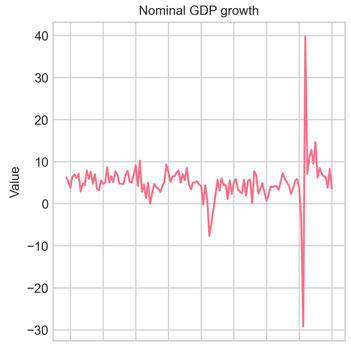
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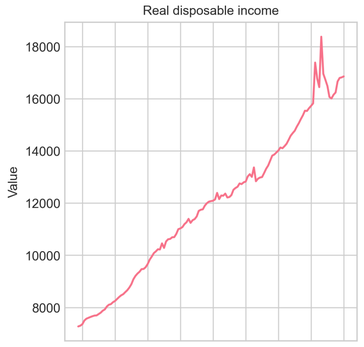
Economic Variables

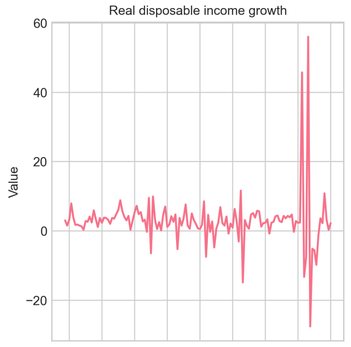


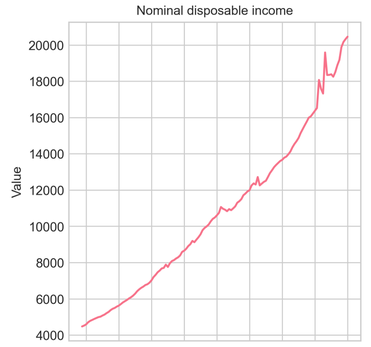


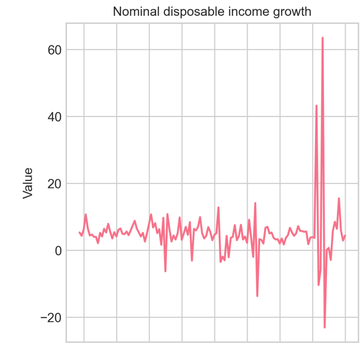


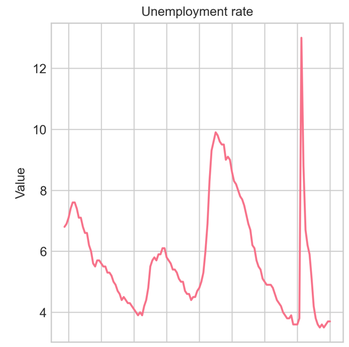


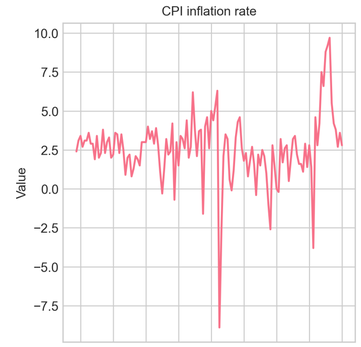






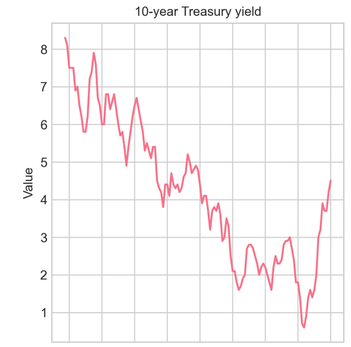






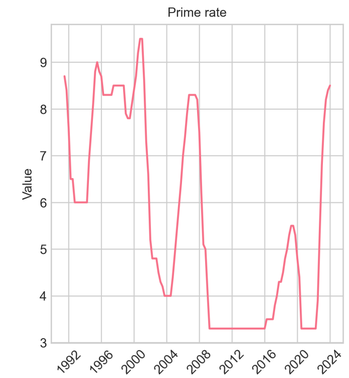


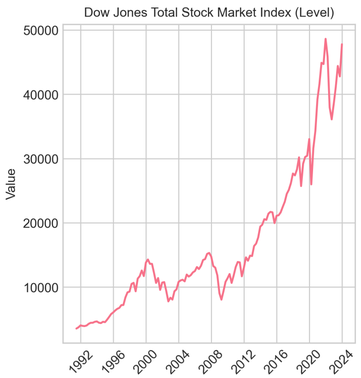


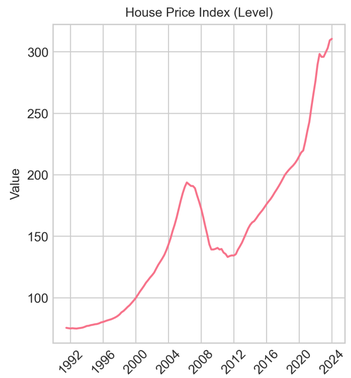


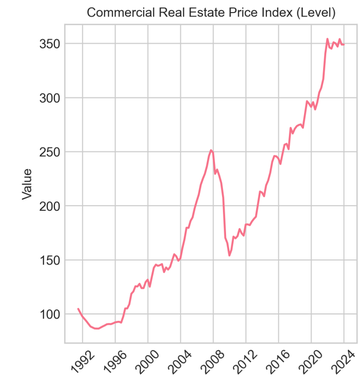


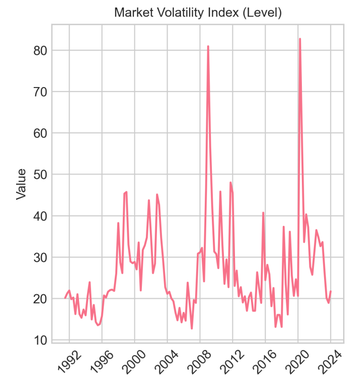


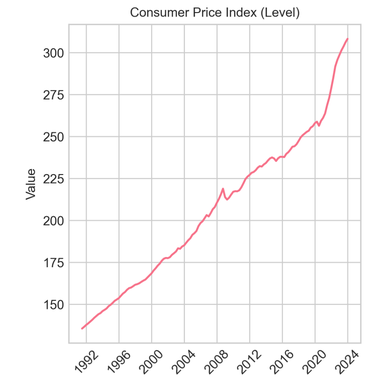












Economic Variables Turn Guesses Into Defendable Forecasts

Economic drivers matter

- Interest rates, unemployment, and GDP growth directly shape deposit inflows and outflows

Behavioral dynamics

- Households hoard cash in recessions or low-rate environments; firms sweep funds in expansions

Limits of pure time series

- A SARIMA without macro variables captures patterns, but not the underlying “why”

Regulatory expectations

- CCAR requires stress-testing deposit forecasts under shocks like rate hikes or recessions

SARIMAX advantage

- Adding exogenous variables (e.g., Fed Funds, inflation) yields defendable forecasts and prevents underestimating deposit runs

Forecasting Edge With Granger Causality For Variable-Selection

Null Hypothesis: X does not Granger-cause Y

Alternative: X provides statistically significant information about future values of Y beyond Y's own past

Decision Rule: Reject H_0 if F-statistic p-value < 0.05

$$Y_t = \alpha + \sum_{i=1}^{p} \beta_i Y_{t-i} + \sum_{i=1}^{q} \gamma_i X_{t-i} + \varepsilon_t$$

Y_t = Deposit variable at time t

α = Intercept term

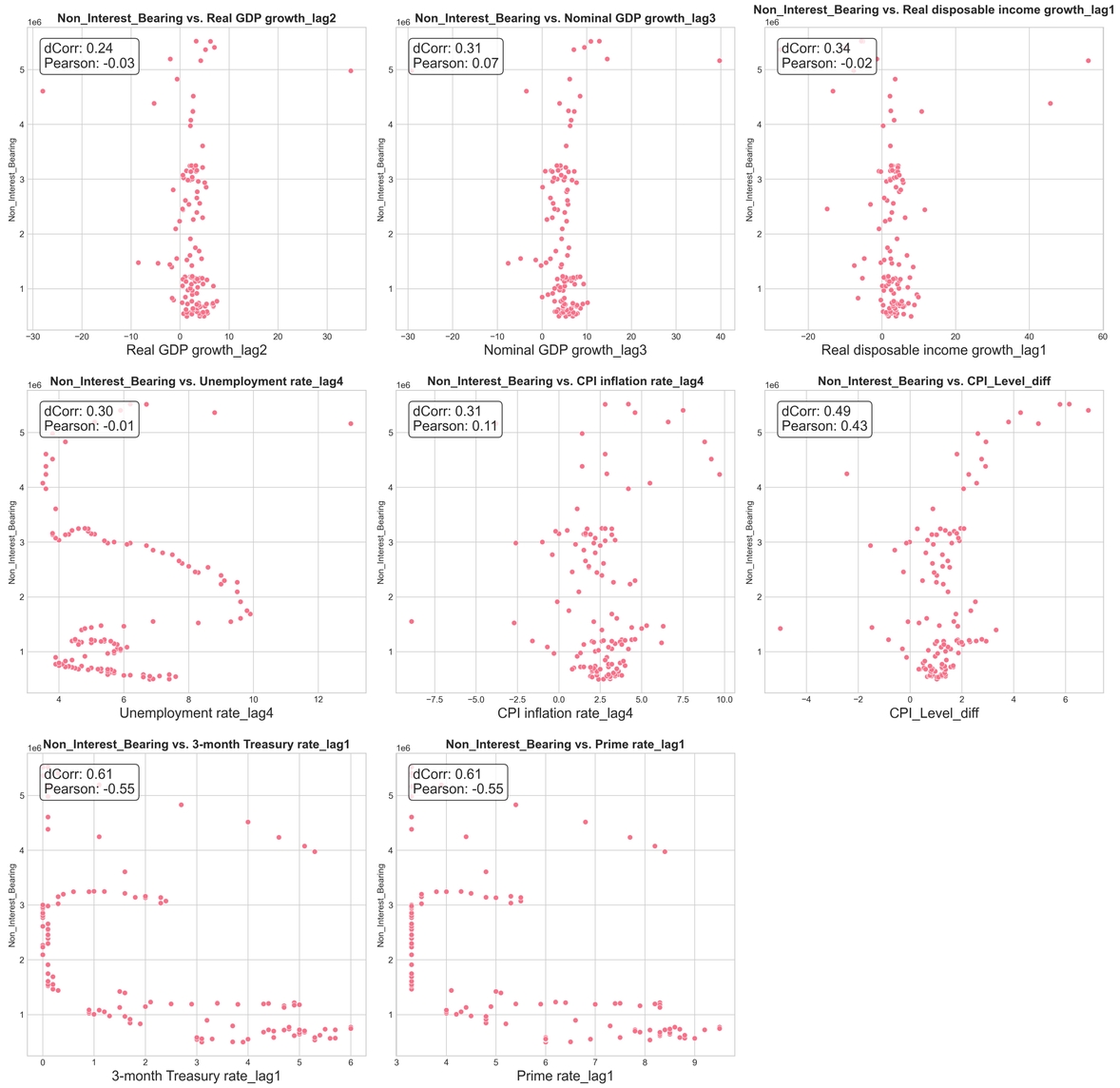
β_i = Coefficients on lagged Y values

γ_i = Coefficients on lagged X values (test variables)

ε_t = Error term

p = Number of lags for dependent variable

q = Number of lags for explanatory variable



Economic Variable	Distance Correlation (dCorr)	Pearson Correlation
3-month Treasury rate (lag 1)	0.61	-0.55
Prime rate (lag 1)	0.61	-0.55
CPI (Level, diff)	0.49	0.43
Real disposable income growth (lag 1)	0.34	-0.02
Nominal GDP growth (lag 3)	0.31	0.07
CPI inflation rate (lag 4)	0.31	0.11
Unemployment rate (lag 4)	0.30	-0.01
Real GDP growth (lag 2)	0.24	-0.03

Non-Interest-Bearing Deposits

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Economic Drivers For Non-Interest-Bearing Deposits

Income & Activity

- Real GDP (lag 2) and nominal GDP (lag 3) drive business cash flows; real disposable income (lag 1) boosts household checking

Labor Market

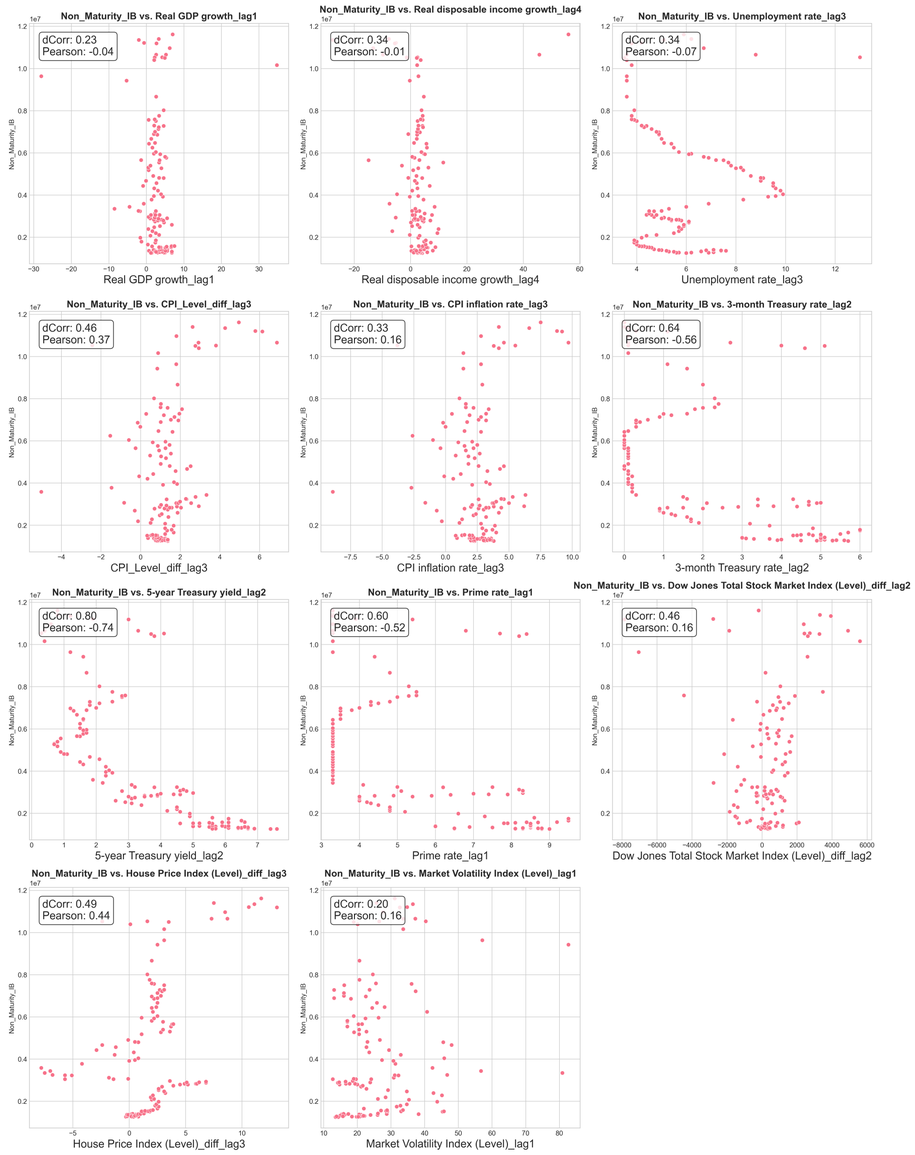
- Unemployment (lag 4) cuts deposit flows as savings deplete, reflecting delayed job loss impact

Inflation Effects

- CPI inflation (lag 4) spurs precautionary cash hoarding; differenced CPI captures immediate price shocks

Interest Rates

- 3-month Treasury (lag 1) and prime rate (lag 1) reduce checking demand as opportunity costs rise



Economic Variable	Distance Correlation (dCorr)	Pearson Correlation
5-year Treasury yield (lag 2)	0.80	-0.74
3-month Treasury rate (lag 2)	0.64	-0.56
Prime rate (lag 1)	0.60	-0.52
House Price Index (Level, diff, lag 3)	0.49	0.44
Dow Jones Stock Market Index (Level, diff, lag 2)	0.46	0.16
CPI Level (diff, lag 3)	0.46	0.37
Real disposable income growth (lag 4)	0.34	-0.01
Unemployment rate (lag 3)	0.34	-0.07
CPI inflation rate (lag 3)	0.33	0.16
Real GDP growth (lag 1)	0.23	-0.04
Market Volatility Index (Level, lag 1)	0.20	0.16

Non-Maturity Interest-Bearing Deposits

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Economic Drivers For Non-Maturity Interest-Bearing Deposits

Income & Growth

- Real GDP (lag 1) fuels immediate savings inflows; real disposable income (lag 4) drives long-term savings accumulation

Uncertainty & Safety

- Unemployment (lag 3) boosts precautionary savings; VIX (lag 1) triggers rapid flight to safe deposits

Inflation Dynamics

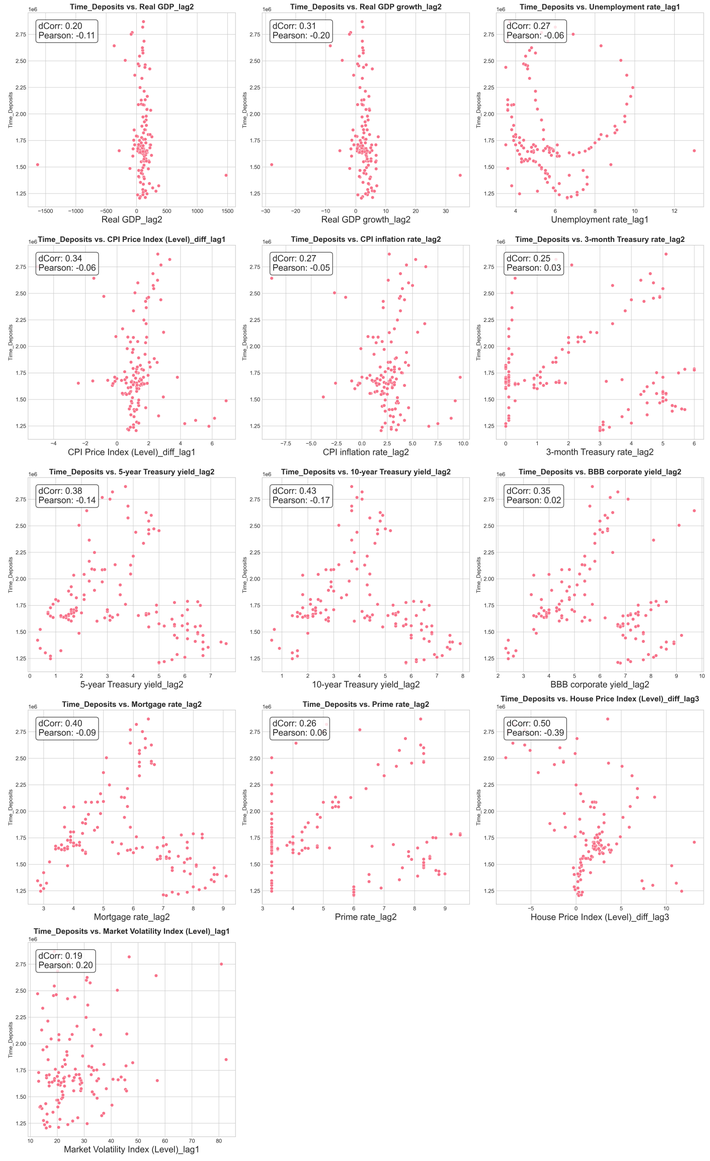
- CPI inflation and differenced CPI (lag 3) shape real return expectations, impacting savings attractiveness

Rates & Yields

- 3-month/5-year Treasury (lag 2) and prime rate (lag 1) drive deposit pricing and inflows via bank repricing

Wealth Effects

- Stock market swings (lag 2) and home price changes (lag 3) drive reallocation to/from savings accounts



Economic Variable	Distance Correlation (dCorr)	Pearson Correlation
House Price Index (Level, diff, lag 3)	0.50	-0.39
10-year Treasury yield (lag 2)	0.43	-0.17
Mortgage rate (lag 2)	0.40	-0.09
5-year Treasury yield (lag 2)	0.38	-0.14
BBB corporate yield (lag 2)	0.35	0.02
CPI Price Index (Level, diff, lag 1)	0.34	-0.06
Real GDP growth (lag 2)	0.31	-0.20
CPI inflation rate (lag 2)	0.27	-0.05
Unemployment rate (lag 1)	0.27	-0.06
Prime rate (lag 2)	0.26	0.06
3-month Treasury rate (lag 2)	0.25	0.03
Real GDP (lag 2)	0.20	-0.11
Market Volatility Index (Level, lag 1)	0.19	0.20

Time Deposits

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Economic Drivers For Time Deposits

Economic Confidence

- Real GDP level/growth (lag 2) boosts long-term CD commitments via stability and growth optimism

Employment & Risk

- Unemployment (lag 1) cuts CD demand due to liquidity fears; VIX (lag 1) drives risk-off flows to safe CDs

Inflation Hedging

- Differenced CPI (lag 1) and CPI inflation (lag 2) spur fixed-rate CDs to lock in returns

Rate Environment

- Treasury (3M, 5Y, 10Y), BBB, mortgage, prime rates (lag 2) shape CD pricing and competitiveness

Wealth Effects

- House price gains (lag 3) fund CDs via refinancing, reflecting delayed wealth reallocation

Deposits Are Chained By Time And Seasonality

SARIMAX for Deposit Modeling

- **Captures Time Dependencies**: Models autocorrelation in deposits (e.g., past balances predict future) via AR and MA terms, unlike static regression
- **Handles Seasonality**: Seasonal AR/MA terms (e.g., SARIMA$(0,0,0)_4$) capture quarterly cycles like tax-driven deposit peaks
- **Adjusts for Trends**: Differencing (d=1, D=1) removes non-stationarity, focusing on quarter-to-quarter changes
- **Incorporates Economic Drivers**: X variables (e.g., GDP, rates) link macro conditions to deposit flows
- **Robust Forecasting**: Balances persistence, shocks, and external factors, ensuring accurate liquidity and capital projections

SARIMAX For Deposit Modeling

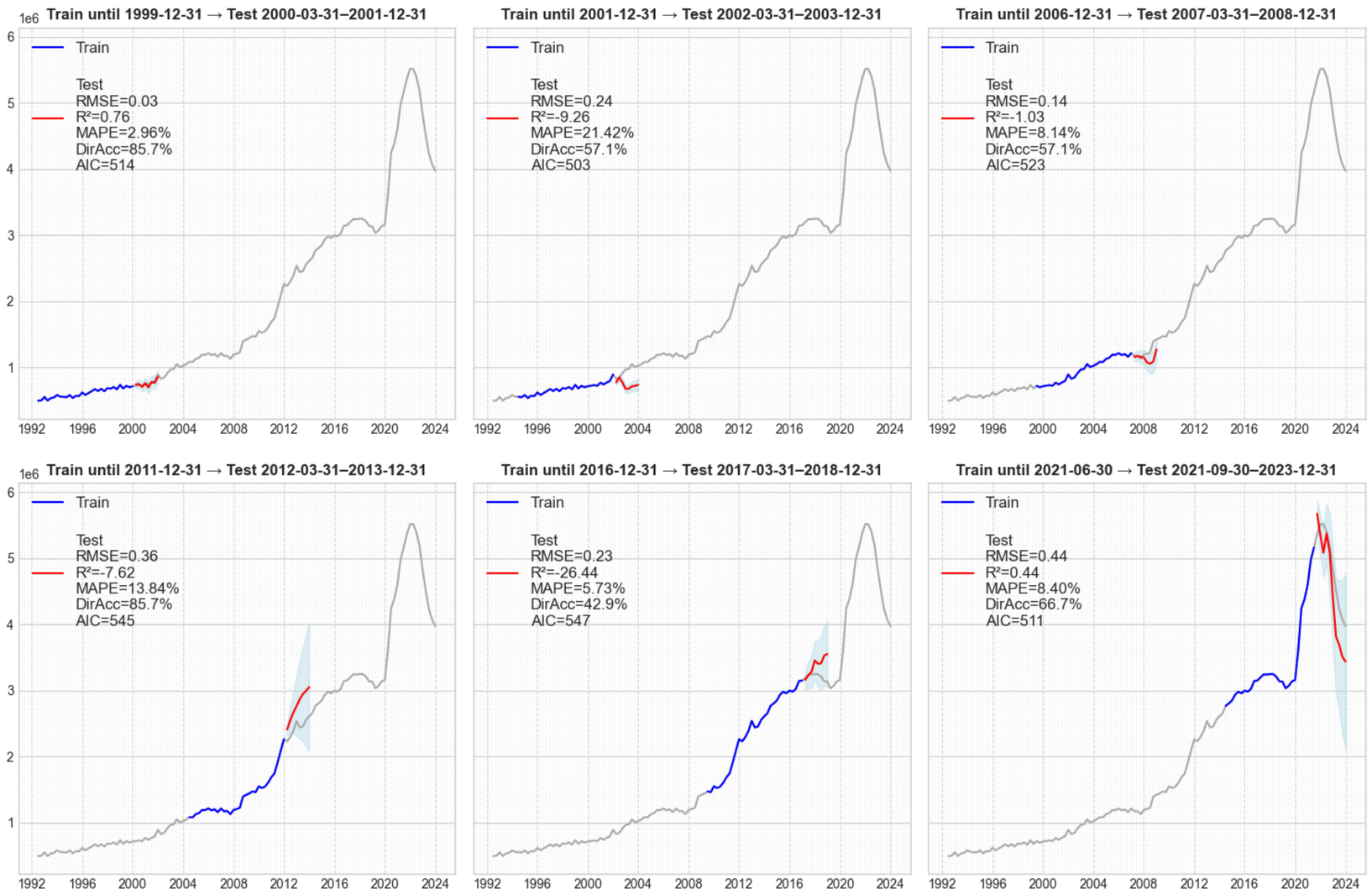
$$Y_t = c + \sum_{i=1}^{p} \phi_i Y_{t-i} + \sum_{i=1}^{q} \theta_i \epsilon_{t-i} + \sum_{s=1}^{P} \Phi_s Y_{t-sS} + \sum_{r=1}^{Q} \Theta_r \epsilon_{t-rS} + \beta X_t + \epsilon_t$$

Term	Description	Economic Meaning
Y_t	Dependent variable at time t	Current deposit level (after differencing)
c	Intercept/constant	Base level or drift term
$\sum_{i=1}^{p} \phi_i Y_{t-i}$	AR(p) component	Past deposit values influence current level (persistence)
$\sum_{i=1}^{q} \theta_i \epsilon_{t-i}$	MA(q) component	Past forecast errors affect current level (shock adjustment)
$\sum_{s=1}^{P} \Phi_s Y_{t-sS}$	Seasonal AR(P)	Seasonal persistence (e.g., Q4 deposits $\rightarrow$ next Q4)
$\sum_{r=1}^{Q} \Theta_r \epsilon_{t-rS}$	Seasonal MA(Q)	Seasonal shock adjustment (e.g., Q4 crisis $\rightarrow$ next Q4)
βX_t	Exogenous variables	Economic drivers (GDP, rates, unemployment, etc.)
ϵ_t	Error term	Random shocks/unexplained variation

Parameters:

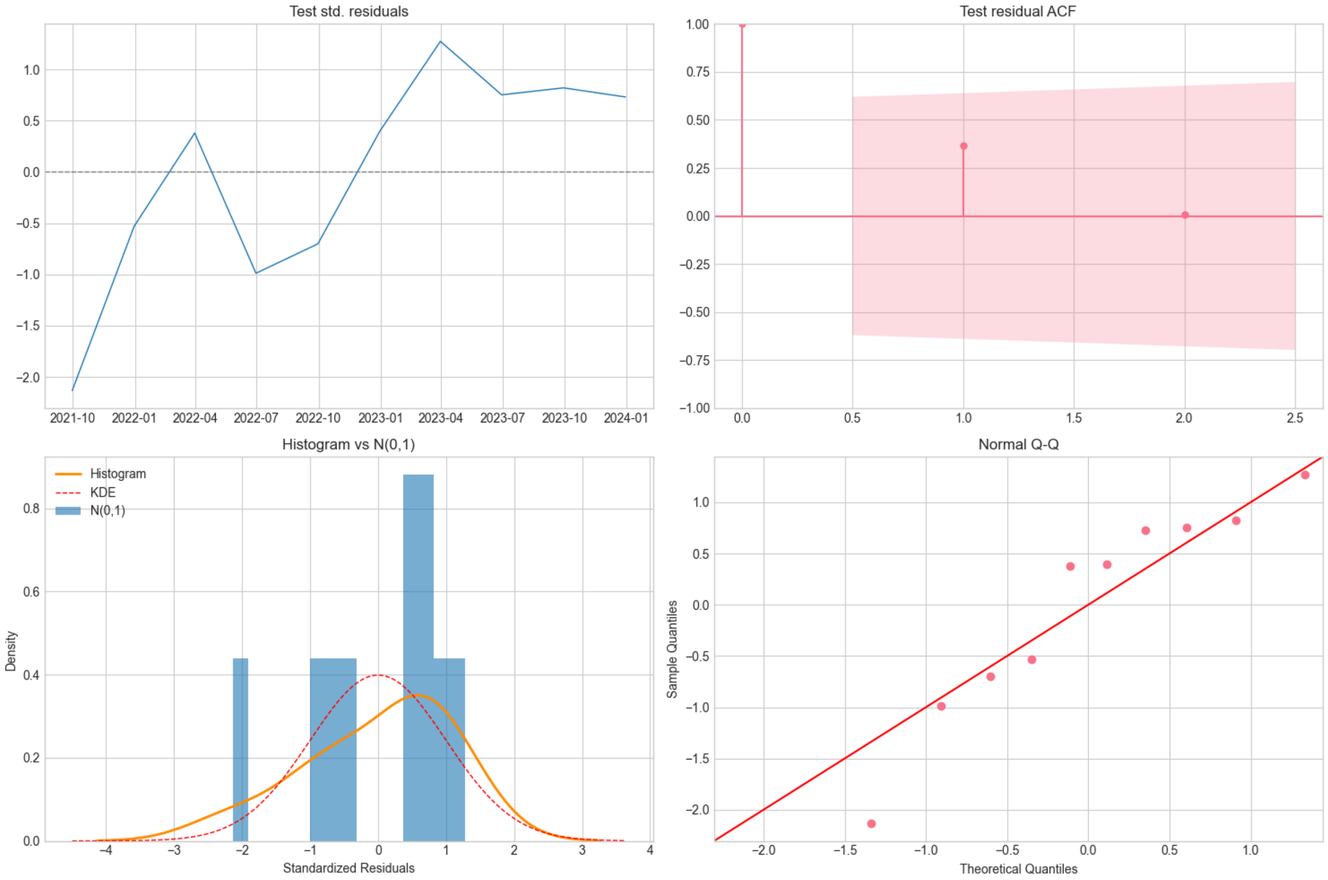
- ϕ_i = Non-seasonal AR coefficients
- θ_i = Non-seasonal MA coefficients
- Φ_s = Seasonal AR coefficients
- Θ_r = Seasonal MA coefficients
- β = Exogenous variable coefficients
- S = Seasonal period (4 for quarterly data)

Non-Interest-Bearing Deposits



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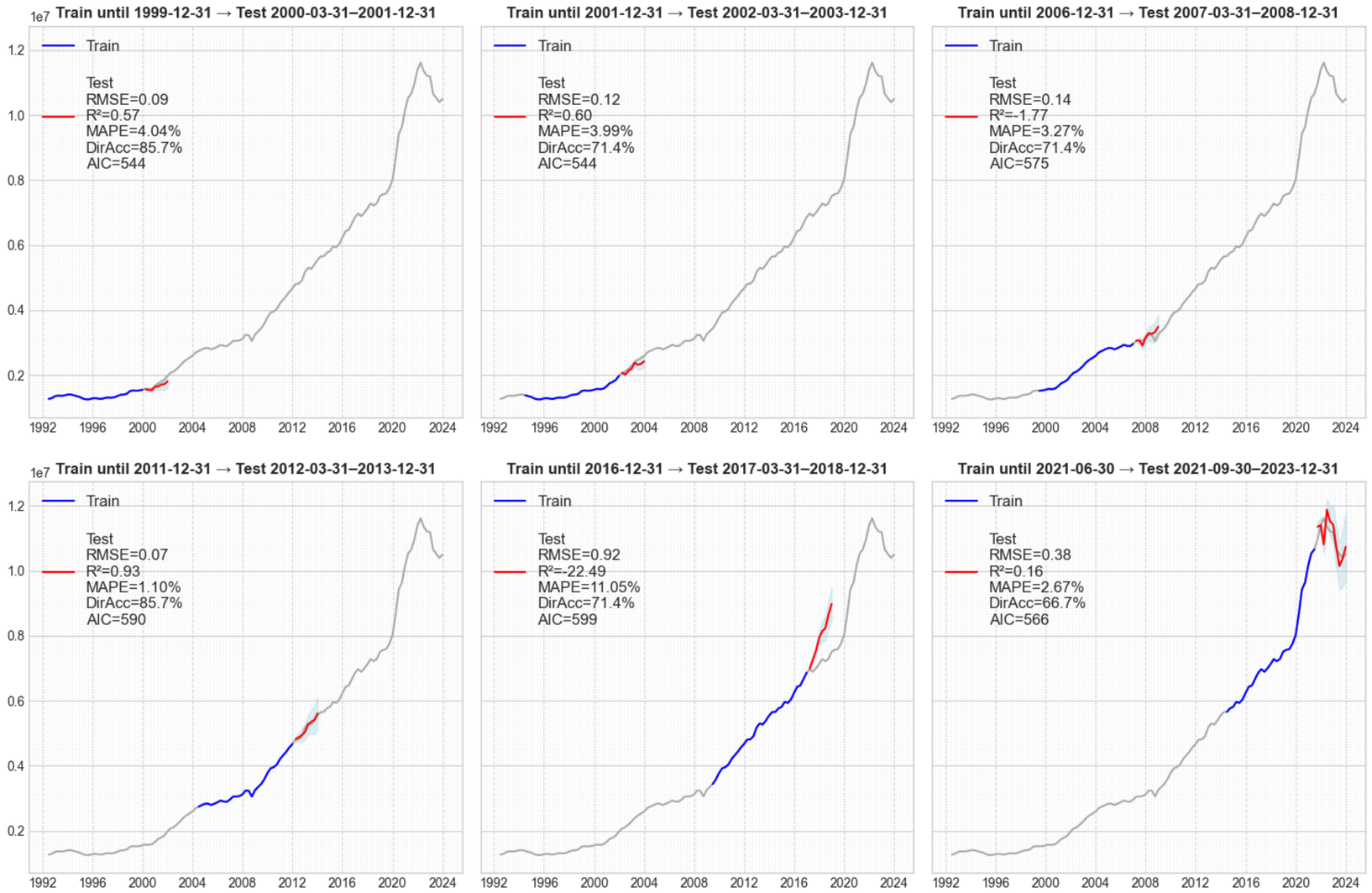
Non-Interest-Bearing Deposits



Test	Statistic	P-value	Conclusion	Status
Ljung-Box (4 lags)	-	0.619	No autocorrelation in residuals	✅ Pass
Jarque-Bera Normality	1.044	0.593	Residuals are normally distributed	✅ Pass

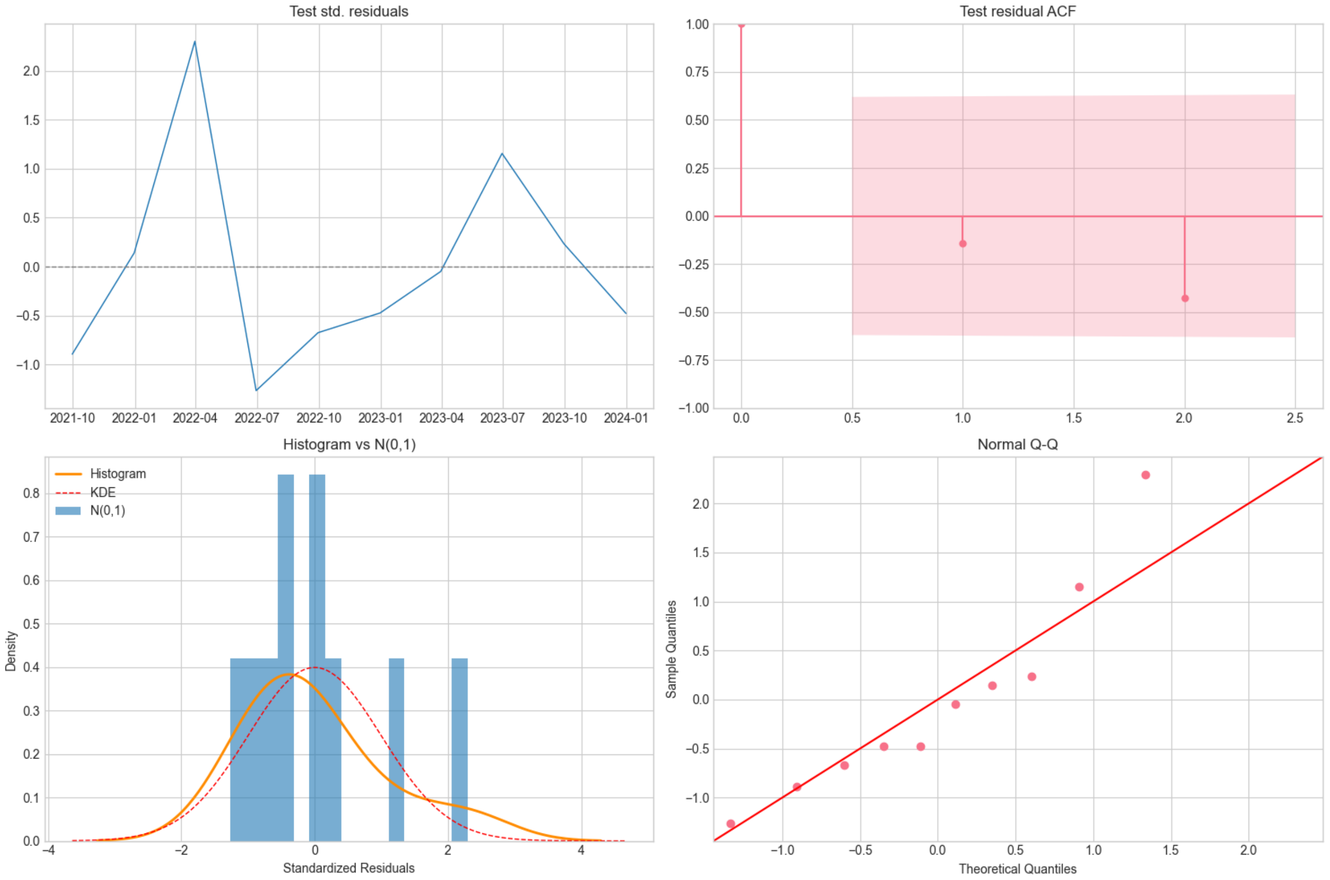
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Non-Maturity Interest-Bearing Deposits



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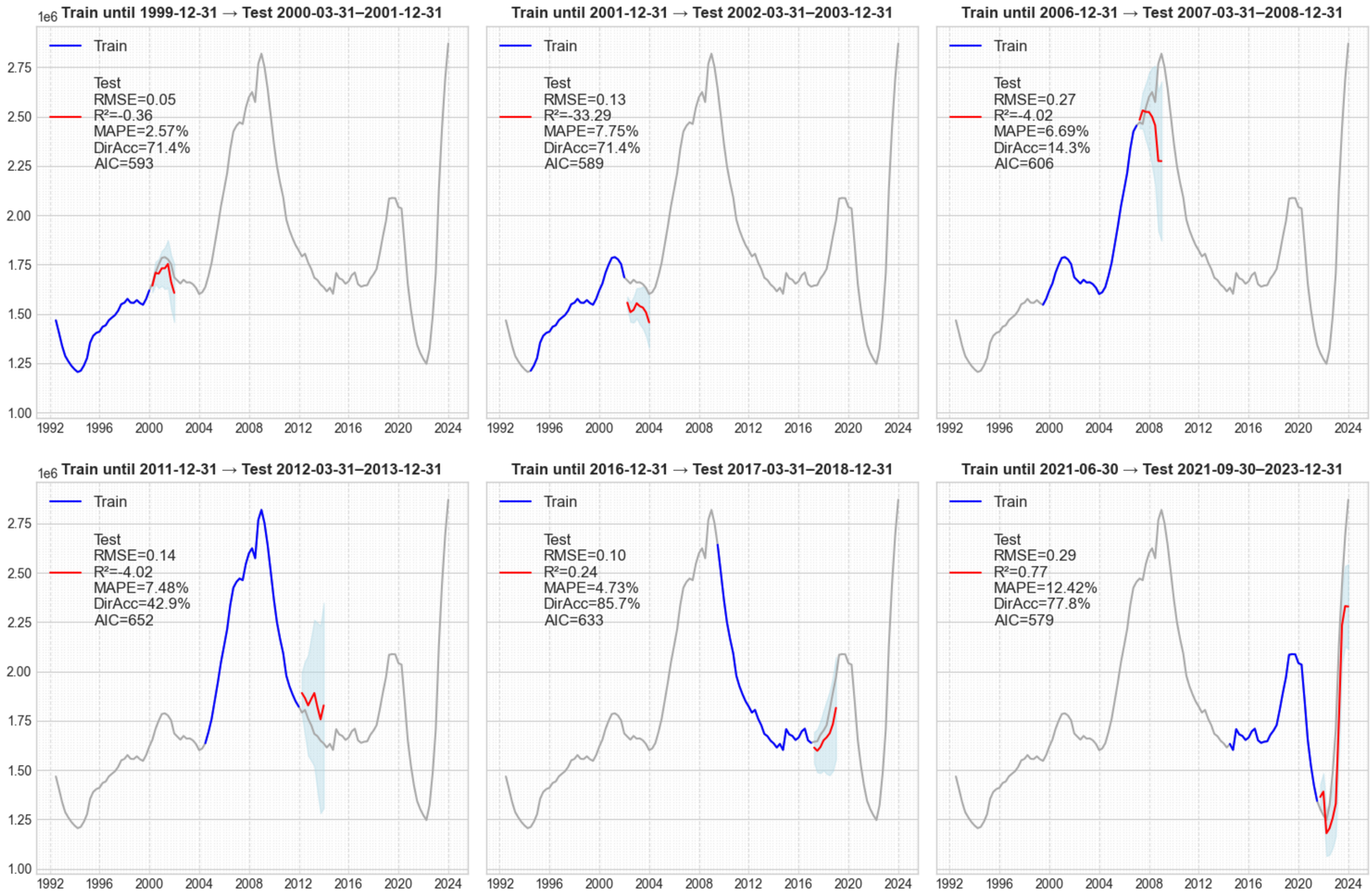
Non-Maturity Interest-Bearing Deposits



Test	Statistic	P-value	Conclusion	Status
Ljung-Box (4 lags)	-	0.509	No autocorrelation in residuals	✅ Pass
Jarque-Bera Normality	1.869	0.393	Residuals are normally distributed	✅ Pass

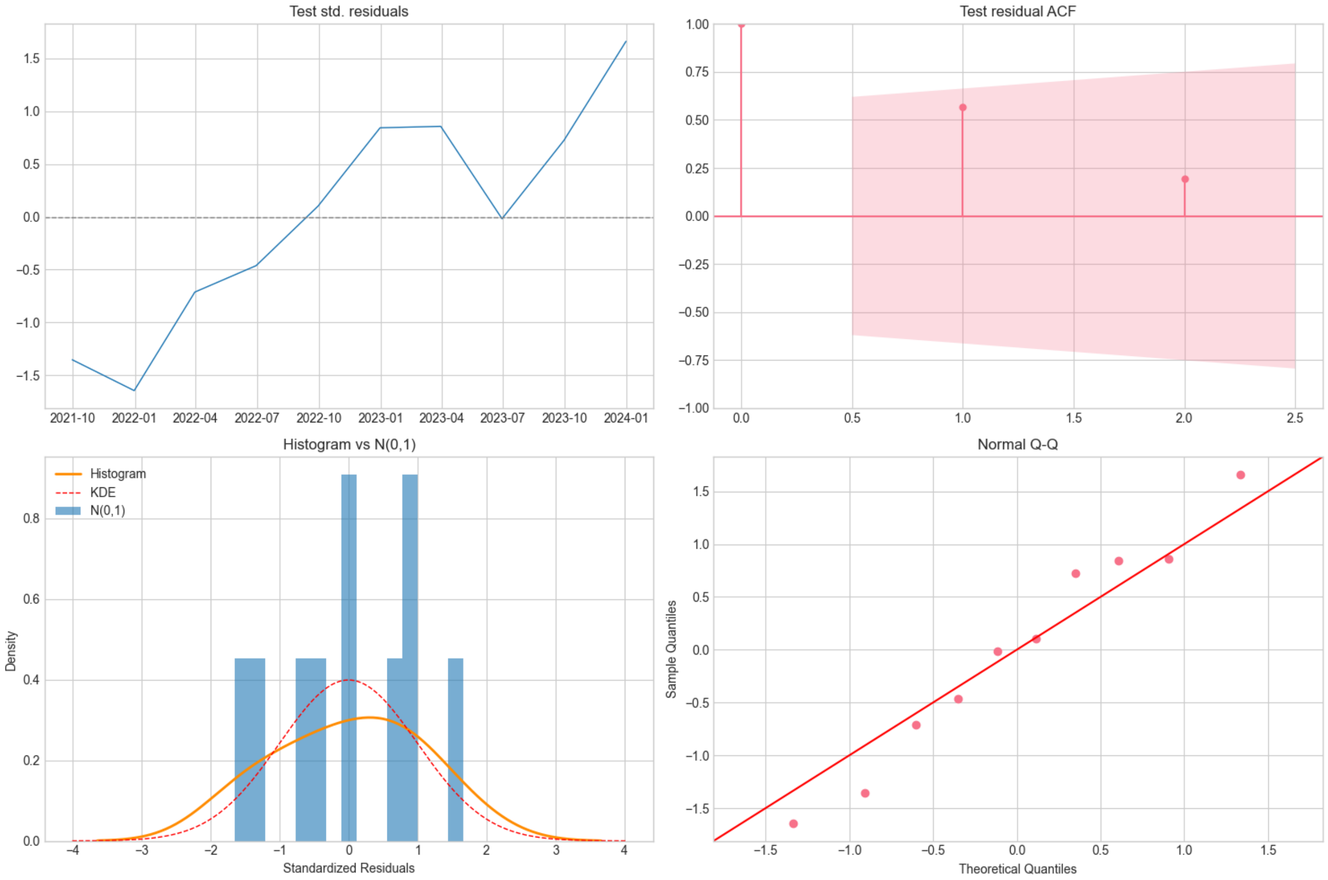
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Time Deposits



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Time Deposits



Test	Statistic	P-value	Conclusion	Status
Ljung-Box (4 lags)	-	0.258	No autocorrelation in residuals	✅ Pass
Jarque-Bera Normality	0.447	0.800	Residuals are normally distributed	✅ Pass

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Deposits Make Up The Bulk Of FDIC's Quarterly Banking Profile

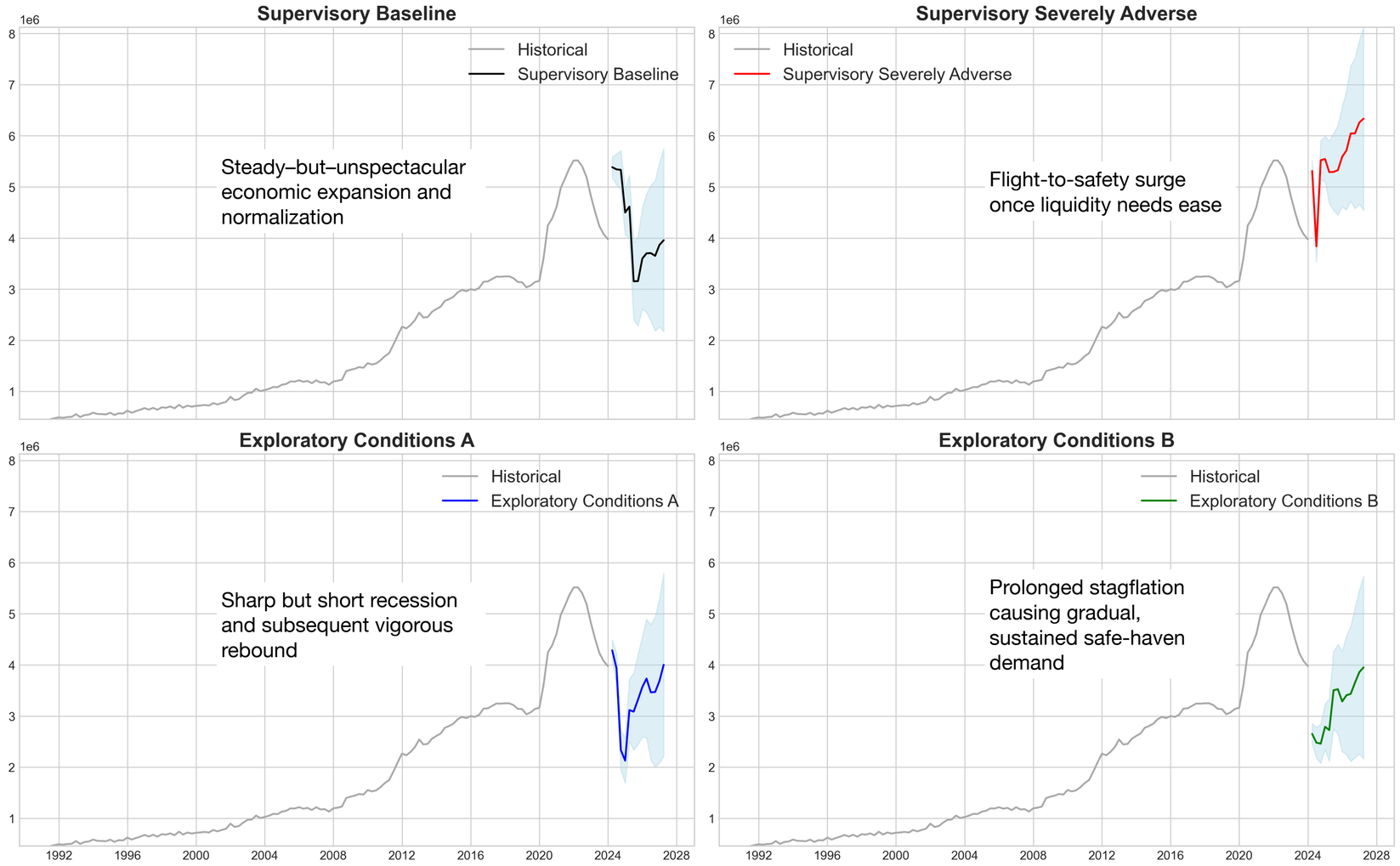
Data from FDIC includes:

Four economic scenarios for forecasting (2024-2027)

- Supervisory Baseline
- Supervisory Severely Adverse
- Exploratory Conditions A
- Exploratory Conditions B

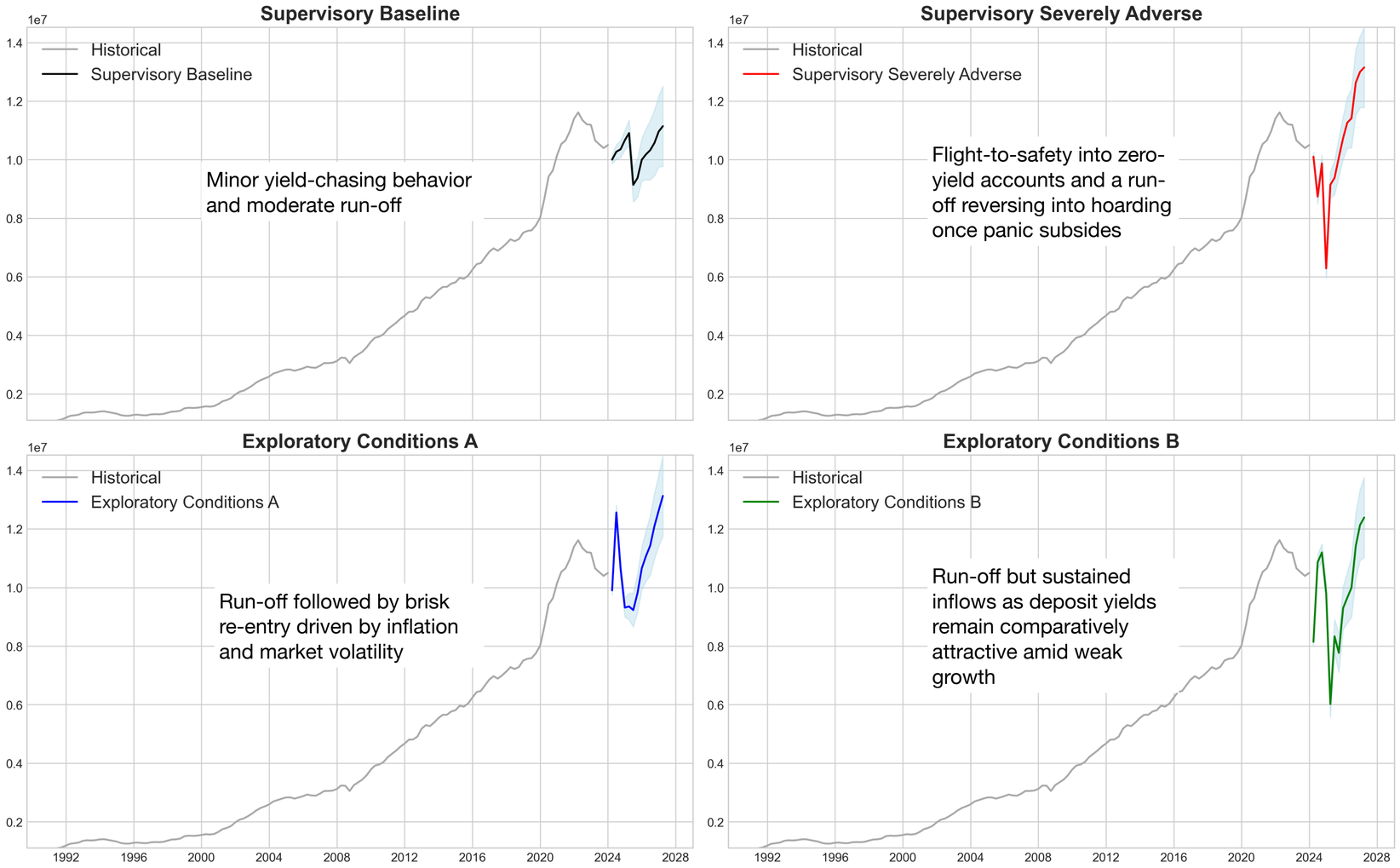
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Non-Interest-Bearing Deposits



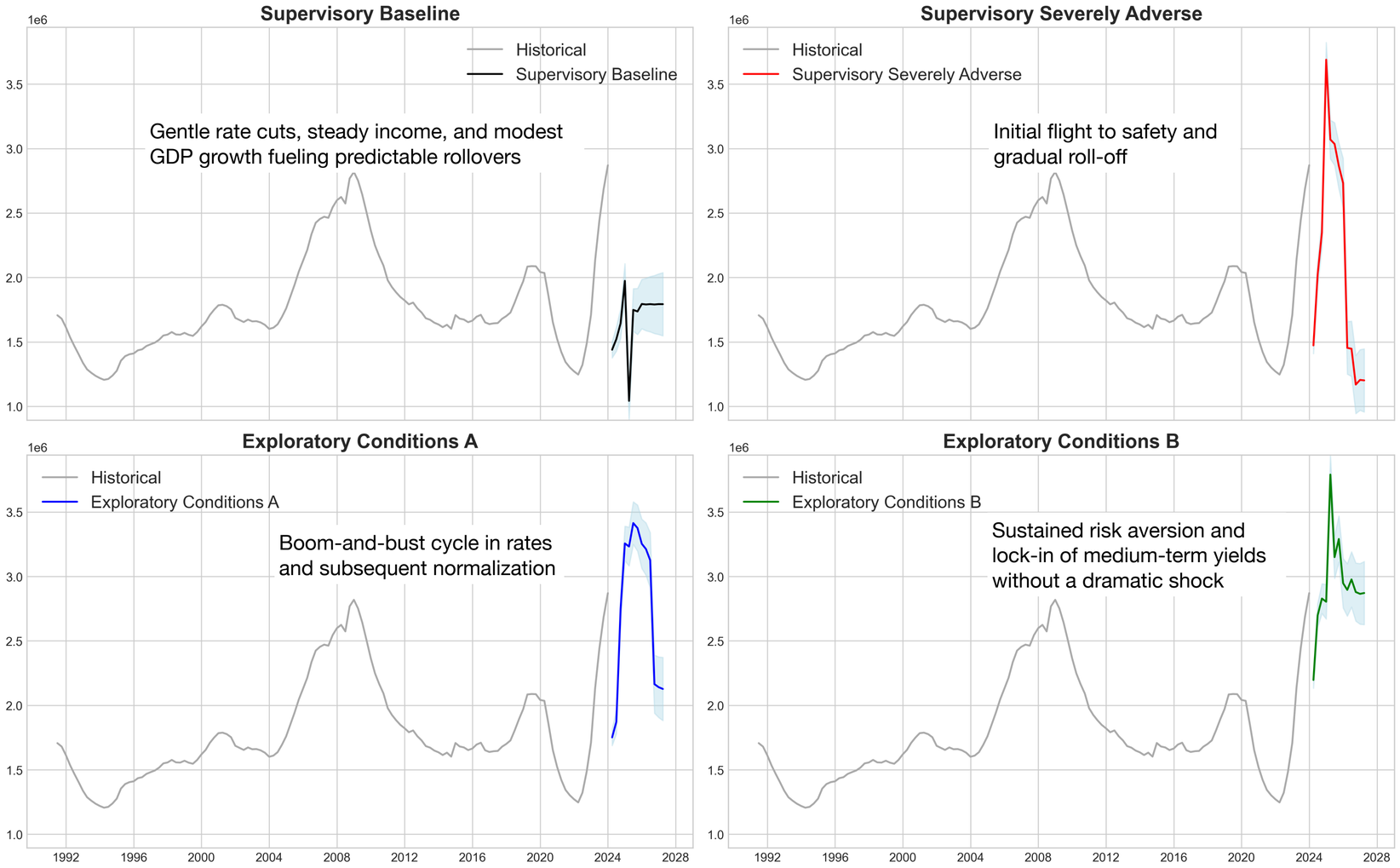
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Non-Maturity Interest-Bearing Deposits



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Time Deposits



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Thank you

	Supervisory Baseline	Supervisory Severely Adverse	Exploratory Conditions A	Exploratory Conditions B
Economic Context	Stable growth, soft landing	Deep 5-quarter recession	Moderate 5-quarter recession	Deep 6-quarter stagflation
GDP Decline	None (0.7% to 2.1% growth)	-11.6% peak contraction	-3.6% peak contraction	-13.2% peak contraction
Unemployment	Low (3.9%-4.3%)	Peaks at 10.0%	Peaks at 6.7%	Peaks at 10.0%, prolonged
Inflation	2.3% (Fed target)	1.3% (disinflation)	3.5% (persistent)	5.0%-6.0% (high, sticky)
Fed Funds Rate	Gradual cuts (5.3% to 3.1%)	Emergency cuts to 0.1%	Constrained (6.2% to 3.3%)	Policy dilemma, high rates
Market Stress	Flat stocks, low VIX (24-28)	-55% stocks, VIX 70	-30% stocks, VIX 50	-55% stocks, VIX 70+
Non-Interest-Bearing Deposits	**Run-off**: 5-10% **Behavior**: Sticky; retail and business maintain balances for transactions. **Driver**: Stable confidence, low unemployment.	**Run-off**: 15-30% **Behavior**: Businesses draw for operations; retail dips for essentials. **Driver**: Fear of bank runs, 10% unemployment.	**Run-off**: 10-25% **Behavior**: Businesses shift to inflation hedges; retail steady. **Driver**: Moderate job losses, sticky inflation.	**Run-off**: 20-40% **Behavior**: Heavy withdrawals for survival. **Driver**: Long recession, eroded trust.
Non-Maturity Interest-Bearing Deposits	**Run-off**: 10-20% **Behavior**: Minor yield-chasing; businesses sweep to investments. **Driver**: Rate cuts reduce appeal.	**Run-off**: 40-60% **Behavior**: Flight to safety (Treasuries) or yields; digital exits spike. **Driver**: Zero rates, VIX spike to 70.	**Run-off**: 30-50% **Behavior**: Businesses chase TIPS; retail seeks higher yields. **Driver**: 3.5% inflation, market volatility.	**Run-off**: 50-70% **Behavior**: Mass exit to inflation-protected assets. **Driver**: 6% inflation, prolonged crisis.
Time Deposits	**Run-off**: <5% **Behavior**: Stable rollovers; risk-averse inflows. **Driver**: Predictable maturities, low volatility.	**Run-off**: 10-20% **Behavior**: Early redemptions rise; maturities not renewed. **Driver**: Liquidity needs, near-zero rates.	**Run-off**: 15-25% **Behavior**: Breakage for inflation hedges; short-term rollovers. **Driver**: High inflation, penalties less deterrent.	**Run-off**: 20-30% **Behavior**: High breakage, low renewals. **Driver**: Stagflation, funding cost pressures.
Key Challenge for Banks	Optimize deposit mix for margins	Survive liquidity drains, bank run risks	Balance inflation-driven outflows	Manage prolonged deposit flight
CCAR Modeling Focus	Low liquidity buffer needs; net interest margin	High HQLA requirements; stress capital adequacy	Model inflation sensitivity; moderate buffers	Large buffers; simulate 1970s-style stagflation
Recovery Pattern	Steady inflows	V-shaped deposit rebound	Slow deposit stabilization	Very slow deposit recovery